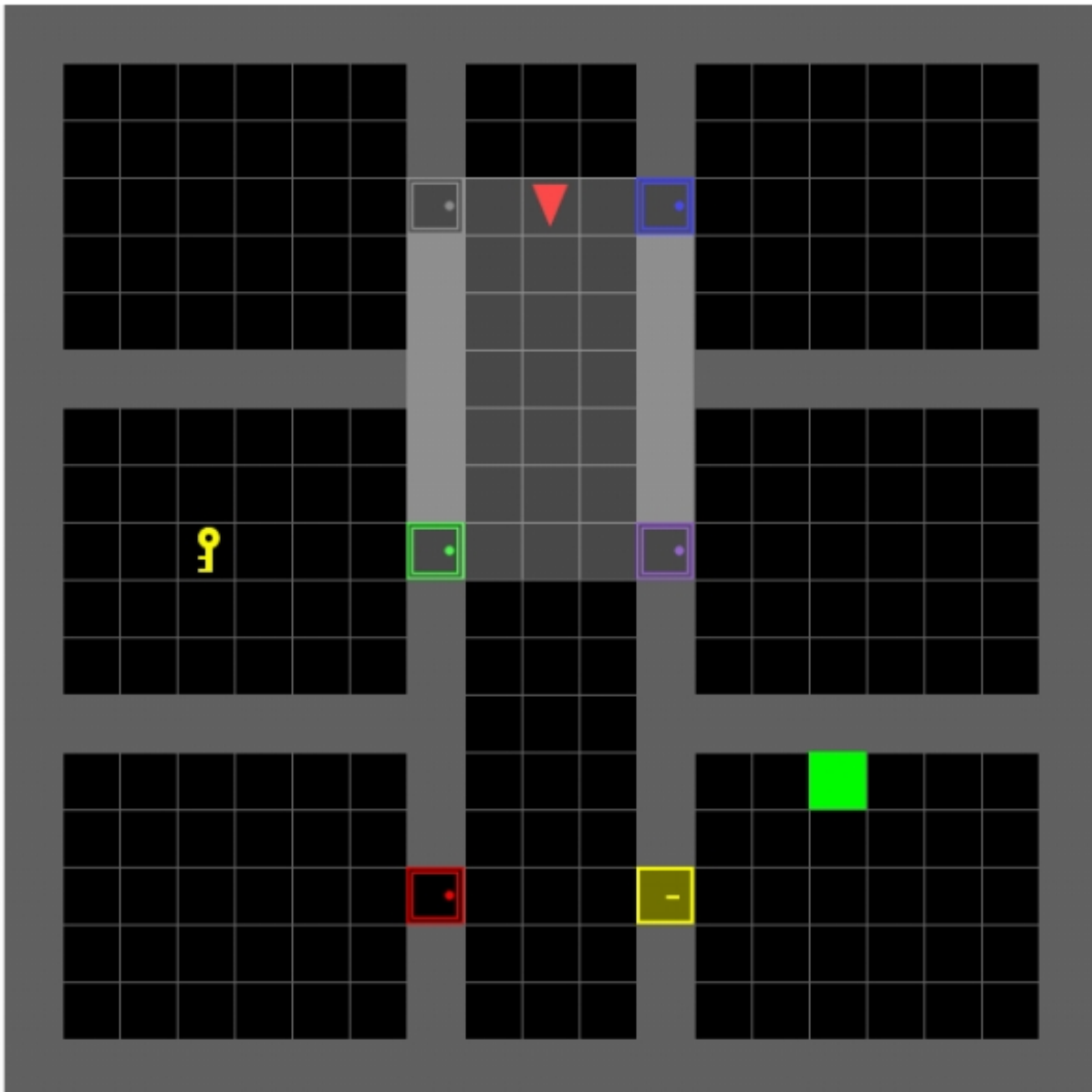


Plan-and-Execute: LLM-Guided Reinforcement Learning with Cross-Modal Fusion for Long-Sequence Decision Making

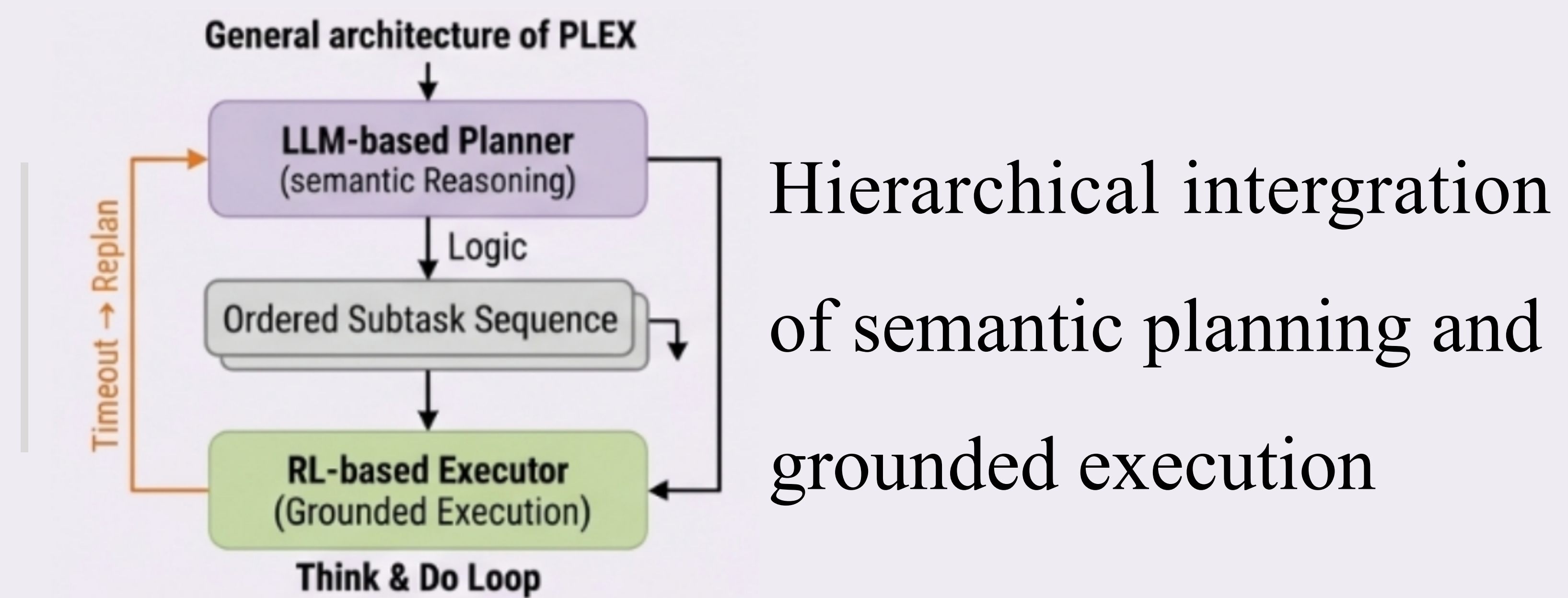
Yadong Li¹, Tong Zhang¹, Zhen Cui²

1. Nanjing University of Science and Technology, 2. Beijing Normal University

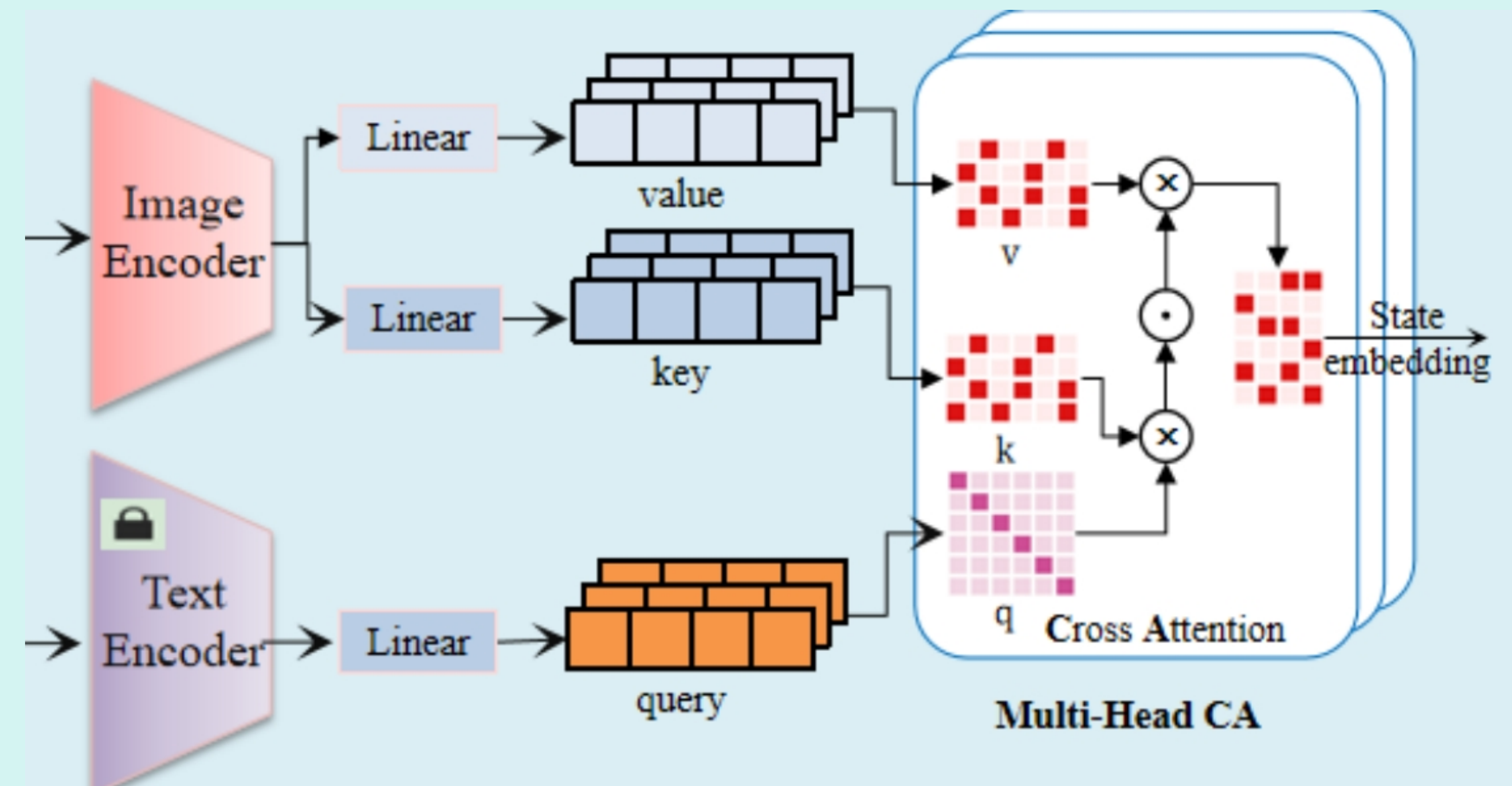
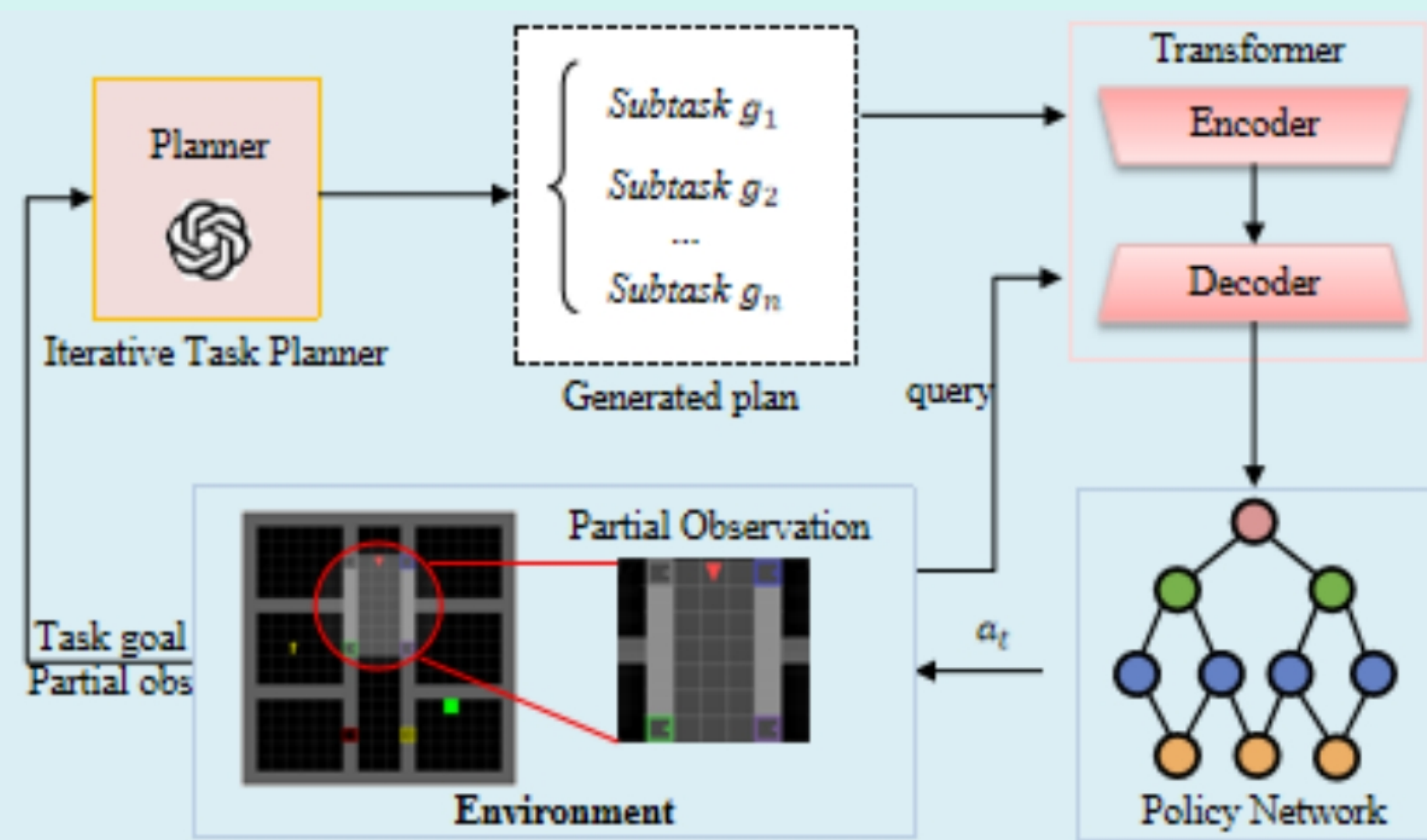
1. MOTIVATION



- Sparse Rewards
- Partial observe
- Changed goals

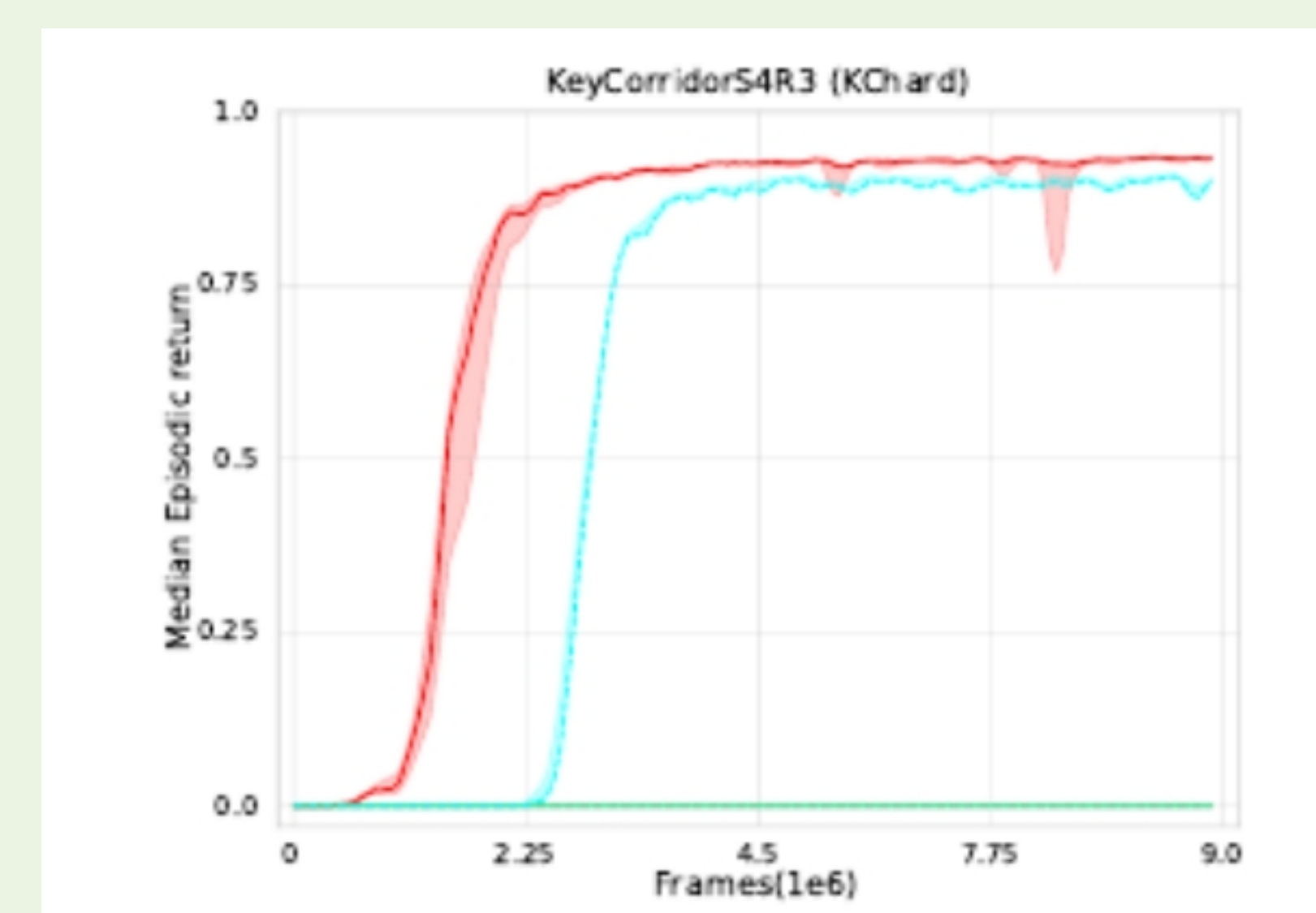
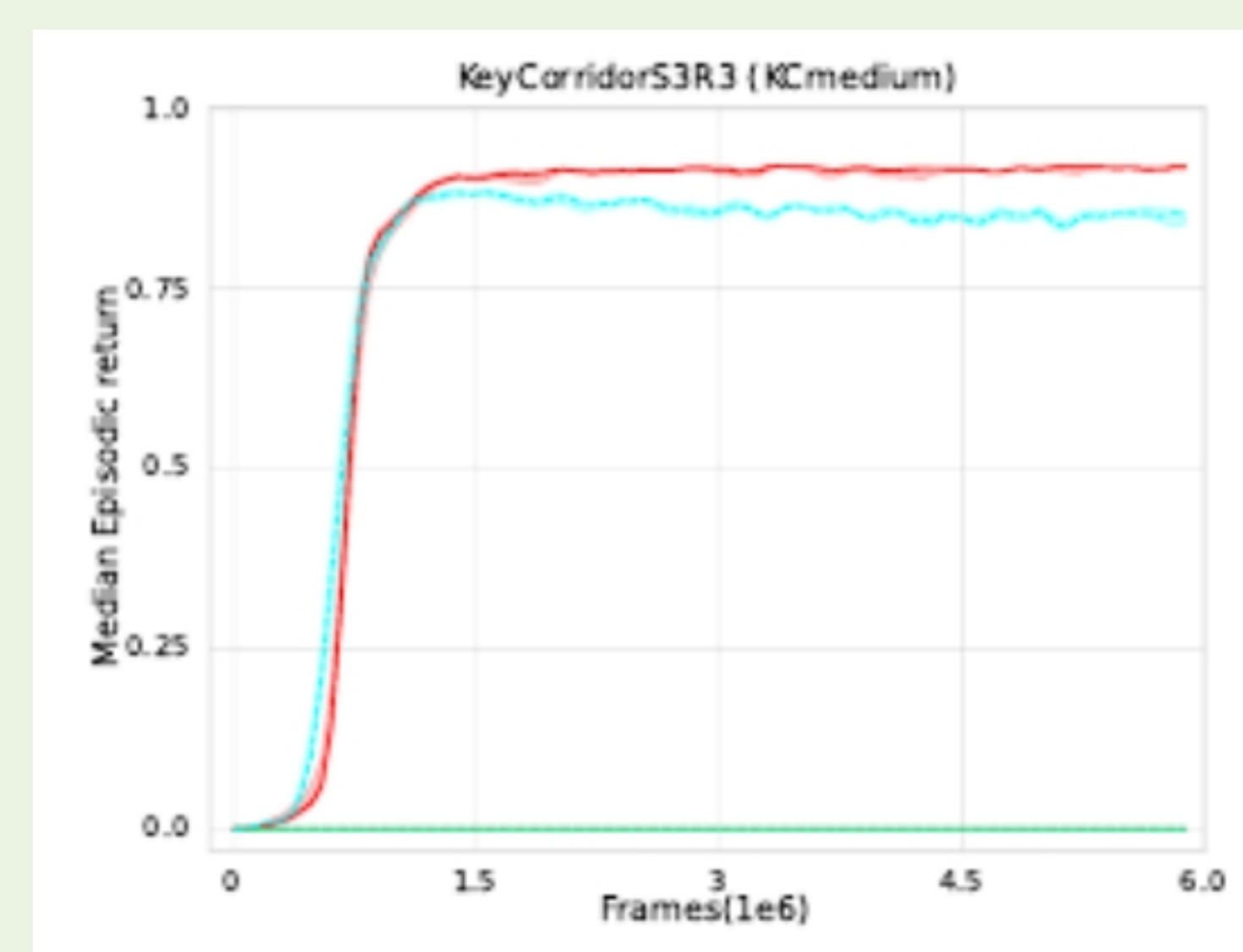
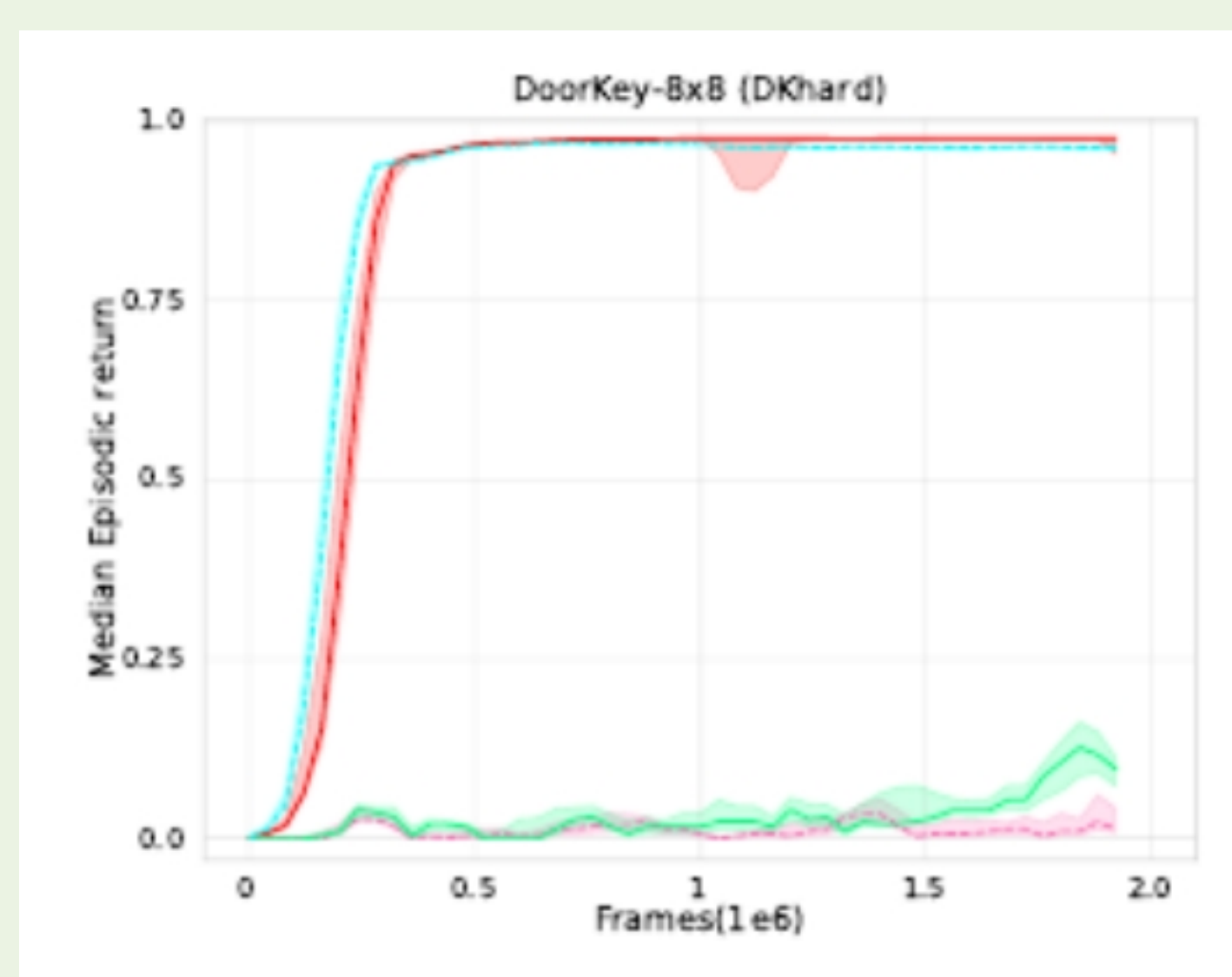
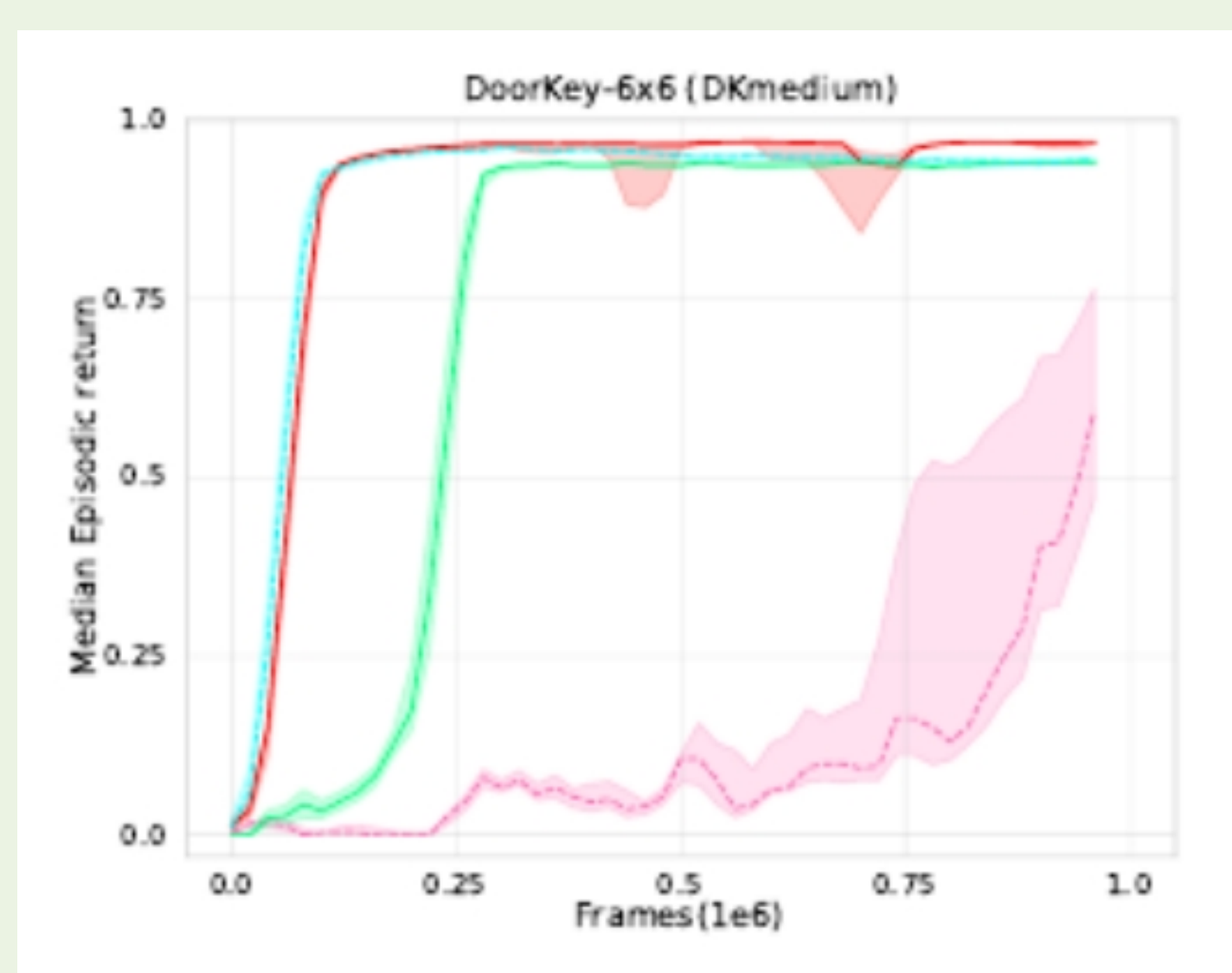


2. METHOD DETAILS



Language-Grounded Reinforcement Learning

3. EXPERIMENTS



4. CONCLUSION

- Hierarchical LLM-RL enhances long-horizon decision making
- Iterative planning provide robust execution
- Ground policy improve context awareness